

Cross-Cohort Generalization Failure in Gut Microbiome-Based Alzheimer's Disease Classifiers: Evidence from Five Independent Cohorts with Batch Correction Analysis

Running title: Microbiome AD classifier generalization failure

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Abstract

Gut microbiome dysbiosis has been proposed as a biomarker for Alzheimer's disease (AD) and mild cognitive impairment (MCI), yet individual studies consistently report contradictory taxonomic signatures across populations, and cross-cohort classifier performance has never been systematically quantified. We assembled five publicly available 16S rRNA gene sequencing cohorts (509 samples; China, Kazakhstan, South Korea; Zhuang 2018, Ling 2020, Zhu 2022, Kaiyrykyzy 2022, Kim 2022) and evaluated whether machine learning classifiers trained within one cohort could generalize to independent cohorts, whether standard batch correction methods could recover transferability, and which taxonomic features drove cohort-specific predictions. DADA2 amplicon sequence variant (ASV) processing and genus-level harmonization produced a unified matrix of 396 genera across all cohorts, transformed to centered log-ratio (CLR) coordinates for compositional analysis. The Zhu 2022 cohort deposited paired blood microbiome and fecal samples; only fecal samples (n=90) were retained for all analyses. Within four labeled cohorts (n=401 for binary AD/CN analyses), logistic regression and LightGBM classifiers achieved within-cohort AUC-ROC values ranging from 0.633 to 0.998 under nested cross-validation, broadly consistent with originally reported performance. In leave-one-cohort-out (LOCO) transfer experiments, LOCO AUC-ROC fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations collapsing to near chance (AUC = 0.504). PERMANOVA on Aitchison distance showed that cohort identity explained 19.3% of total compositional variance (F=30.13, $p<0.0001$, 9,999 permutations) versus only 1.4% for disease label—an approximately 12-fold disparity that mechanistically explains why batch correction cannot recover generalization: both ComBat-seq and MMUPHin worsened rather than improved LOCO AUC across all conditions. SHAP analysis revealed near-zero pairwise Jaccard similarity between cohorts' top-20 predictive taxa (mean 0.101 for logistic regression; 0.026 for LightGBM) and ten directional flip taxa for logistic regression and three for LightGBM—genera whose association with AD versus cognitive normality reverses sign across cohorts, including *Akkermansia* (AD-associated in the Ling 2020 cohort by LightGBM; CN-associated in all three others) and *Lactobacillus* (AD-

associated in Kazakhstan; CN-associated in Chinese cohorts). These results demonstrate that gut microbiome-based AD classifiers trained in one population do not transfer reliably to other populations, that the cohort effect is structurally much larger than the disease effect, and that linear batch correction methods are insufficient to bridge this gap. Multi-cohort harmonized training data and prospective validation across geographically and demographically diverse populations are prerequisites before microbiome-based AD classifiers can be considered clinically actionable.

1. Introduction

Alzheimer's disease (AD) is the leading cause of dementia worldwide, affecting an estimated 55 million people, with projections exceeding 150 million by 2050 (World Health Organization, 2023). The gut–brain axis has emerged as a mechanistic pathway linking intestinal microbial communities to neurodegeneration: altered gut microbiome composition has been associated with AD pathology through neuroinflammatory, metabolic, and neuroendocrine routes (Cryan et al., 2019; Kowalski and Mulak, 2019). Since the landmark observation that germ-free mice colonized with gut microbiota from AD patients exhibited greater amyloid- β deposition than those colonized with microbiota from healthy donors (Harach et al., 2017), dozens of 16S rRNA and shotgun metagenomic studies have reported significant differences in gut microbial composition between AD or MCI patients and cognitively normal (CN) controls.

The clinical appeal of a gut microbiome-based diagnostic signature is considerable: stool is non-invasive, inexpensive to collect, and microbiome profiling is increasingly accessible. Multiple independent research groups have trained machine learning classifiers on gut microbiome abundance profiles and reported promising within-cohort discrimination performance, with area under the receiver operating characteristic curve (AUC-ROC) values of 0.70–0.90 in several published studies (Zhuang et al., 2018; Ling et al., 2021; Zhu et al., 2022). However, a critical property of any diagnostic biomarker—transferability across independent populations—has rarely been tested in this literature.

The limited cross-cohort comparisons that do exist reveal a striking inconsistency in the direction of reported associations. A systematic review and meta-analysis synthesizing 17 published studies (679 AD/MCI patients, 632 controls) documented that *Bacteroides* abundance was significantly elevated in North American cohorts yet significantly reduced in Chinese cohorts—a directional reversal across geographically distinct populations (Jemimah et al., 2023). Similar contradictions have been observed for *Akkermansia*, *Faecalibacterium*, and *Lactobacillus* across individual studies (Vogt et al., 2017; Liu et al., 2019; Zhuang et al., 2018). These population-level contradictions suggest that published within-cohort classifiers may be detecting cohort-specific compositional signatures—driven by diet, geography, host genetics, sequencing protocols, or a combination thereof—rather than a universal microbial correlate of AD pathology.

Despite this recognized heterogeneity, no study has systematically quantified: (1) the magnitude of cross-cohort AUC collapse when trained classifiers are applied to independent cohorts; (2) the relative contribution of technical batch effects versus biological heterogeneity to this collapse, as estimated by variance partitioning; and (3) whether state-of-the-art microbiome batch correction

methods can recover cross-cohort generalization. Answering these three questions jointly in one rigorous analysis is the primary contribution of this paper.

We assembled five publicly available 16S rRNA gene sequencing cohorts spanning three countries (China, Kazakhstan, South Korea) and conducted a comprehensive cross-cohort generalization evaluation. We report that: classifiers trained within one cohort consistently fail to generalize to other cohorts; cohort identity explains approximately 12 times more compositional variance than disease status; standard batch correction methods (ComBat-seq and MMUPHin) worsen rather than improve cross-cohort AUC; and the specific taxa driving predictions are almost entirely cohort-specific and frequently contradict in direction across populations. Together, these findings constitute a systematic quantification of the generalization failure in the gut microbiome-based AD classifier literature, with direct implications for the field's trajectory toward clinical translation.

2. Materials and Methods

2.1 Cohort Identification and Data Acquisition

We performed a systematic search of NCBI SRA and ENA for publicly available 16S rRNA amplicon sequencing studies of gut microbiota in AD, MCI, or cognitively normal populations, deposited before January 2025. Studies were included if they provided: (1) per-sample diagnosis labels (AD, MCI, or CN) accessible in public metadata; (2) raw FASTQ files or quality-filtered reads deposited in a public archive; (3) minimum sample sizes of 40 participants per cohort. We restricted this analysis to 16S rRNA studies; shotgun metagenomic cohorts were identified but excluded in this version to avoid the additional platform confound introduced by mixing amplicon and whole-genome sequencing data.

Five cohorts meeting these criteria were included. The ALBION 2025 Greek cohort (PRJNA1297934) was identified but excluded because data had not yet been publicly released at the time of analysis. A sixth cohort (Kim/KBASE 2022, PRJEB50447, South Korea) was included in unsupervised analyses only; per-sample amyloid-PET diagnosis labels are IRB-restricted and not deposited in any public archive (see Supplementary Note S1 for complete documentation of metadata search). Group-level counts from the published paper (18 amyloid-positive preclinical AD; 60 cognitively normal) were available but insufficient for supervised classifier training or evaluation.

The four cohorts with per-sample diagnosis labels used in supervised analyses were:

Zhuang 2018 (China; PRJNA554111): 86 participants (43 AD, 43 CN) from the Third Military Medical University, Chongqing. Paired-end MiSeq 2×300 bp, V3–V4 region (Zhuang et al., 2018).

Ling 2020 (China; PRJNA633959): 171 participants (100 AD, 71 CN). Paired-end MiSeq 2×300 bp, V3–V4 region (Ling et al., 2021).

Zhu 2022 (China; PRJNA489760): 180 samples deposited in SRA from a larger 302-participant study (60 AD, 60 MCI, 60 CN × 2 sample types = 180). Each participant contributed

two samples: a fecal (F_{-}) *sample and a blood microbiome* (B_{-}) sample. Only the 90 fecal samples (30 AD, 30 MCI, 30 CN) were used in all analyses; the 90 blood microbiome samples were excluded as they represent a distinct biological compartment and would confound within-cohort classifiers. Binary analyses used 30 AD and 30 CN fecal samples ($n=60$). Paired-end MiSeq 2×300 bp, V3–V4 (Zhu et al., 2022).

Kazakhstan 2022 (PRJNA811324): 84 participants (41 AD, 43 CN). Single-end NovaSeq sequencing, V3–V4 region (Kaiyrlykzy et al., 2022). This cohort is the only single-end dataset in the study; implications for read length and taxonomic resolution are discussed in Section 2.2 and the Limitations section.

The Kim/KBASE 2022 cohort (Jung et al., 2022; PRJEB50447; 78 participants; 18 preclinical AD by amyloid-PET, 60 CN) was included in PERMANOVA analyses (Phase 4) only and is explicitly excluded from all supervised classifier training and evaluation.

All raw FASTQ data were downloaded from NCBI SRA using fasterq-dump (SRA Toolkit v3.0.10). Cohort-level sequencing statistics are reported in Supplementary Table S2.

2.2 DADA2 Amplicon Sequence Variant Processing

Raw paired-end reads (four cohorts) and single-end reads (Kazakhstan) were processed independently per cohort using DADA2 v1.28 in R v4.3 (Callahan et al., 2016), to avoid introducing cross-cohort data leakage during error model estimation.

For paired-end cohorts, reads were trimmed to truncation lengths selected per cohort based on per-base quality score profiles (generally 240 bp forward / 200 bp reverse for V3–V4 MiSeq 2×300 data), filtered to remove reads with >2 expected errors, chimeras were removed using the consensus method, and forward and reverse reads were merged. For Kazakhstan (single-end), the single-end DADA2 workflow was applied without merging; truncation length was adjusted to 250 bp after inspection of quality profiles (see Supplementary Note S2 for rationale). The mixed single-end/paired-end design is an additional technical source of between-cohort heterogeneity acknowledged as a study limitation.

Taxonomic assignment used the SILVA reference database v138.1. ASVs were collapsed to genus level by summing raw counts per sample. A genus was retained across the entire study if it was observed at ≥1 count in ≥2 samples from at least one cohort; this threshold was applied per cohort before cross-cohort union. The union of retained genera across all five cohorts produced a unified feature set of 396 genera. Genera absent in a given sample received a count of zero.

2.3 Compositional Data Transformation

Raw genus-level count tables were converted to centered log-ratio (CLR) coordinates for all downstream analyses requiring distributional assumptions on unconstrained data (machine learning classifiers, PERMANOVA on Euclidean distance). The CLR transformation is defined as:

$$\text{clr}(x_i) = \log\left(\frac{x_i}{g(x)}\right)$$

where $g(x)$ is the geometric mean of the composition x .

Zero counts were replaced prior to transformation using the multiplicative replacement strategy with $\delta=0.65$:

$$x_i^* = \begin{cases} \delta / p & \text{if } x_i = 0 \\ (1 - n_0 \cdot \delta / p) \cdot x_i & \text{if } x_i > 0 \end{cases}$$

where p is the total number of components and n_0 is the count of zeros in the sample. This strategy preserves the ratios among non-zero components while introducing a minimal perturbation proportional to the relative frequency of zeros, and is preferable to the commonly used additive pseudocount which distorts low-abundance taxa more severely.

CLR-transformed values were used as features for all machine learning models and for Aitchison distance computation (Euclidean distance on CLR values). Relative abundances (proportions, no CLR) were used as input to ComBat-seq (which requires non-negative input) and for Bray-Curtis distance.

2.4 Within-Cohort Classifier Baselines

We trained binary classifiers (AD vs. CN) independently for each of the four labeled cohorts using two model families:

Logistic Regression with Elastic Net regularization (scikit-learn v1.5; Pedregosa et al., 2011; LogisticRegression, solver="saga", L1-ratio=0.5). Regularization strength C was tuned over {0.01, 0.1, 1.0, 10.0} in inner cross-validation.

LightGBM gradient-boosted trees (lightgbm v4; Ke et al., 2017). Hyperparameters tuned: number of estimators $\in \{100, 200, 500\}$, learning rate $\in \{0.05, 0.1, 0.2\}$, max depth $\in \{3, 5, -1\}$.

For Zhu 2022, AD was defined as the positive class and MCI samples were excluded from binary analyses (resulting in $n=60$: 30 AD, 30 CN fecal samples); a separate three-class analysis (AD / MCI / CN) used all 90 fecal samples and is reported as a secondary finding. Because only fecal samples were retained, each patient appears exactly once; standard StratifiedKFold was used without any paired-sample grouping structure.

Performance was estimated using nested cross-validation: a 10-fold outer loop for performance estimation and a 5-fold inner loop for hyperparameter tuning. The primary metric was AUC-ROC computed on out-of-fold predictions pooled across the 10 outer folds, providing a single unbiased AUC estimate per cohort-model combination. Bootstrap 95% confidence intervals (1,000 resamples) on the pooled out-of-fold predictions are reported throughout. We also report sensitivity and specificity at the Youden's J optimal threshold.

2.5 Cross-Cohort Generalization Experiment

We evaluated two cross-cohort transfer designs:

Leave-one-cohort-out (LOCO): For each test cohort, a classifier was trained on the pooled samples from the remaining three labeled cohorts (no cross-validation within this training step; hyperparameters were selected via 5-fold inner CV on the training pool). The trained model was

then evaluated on the held-out cohort. This produced four LOCO AUC estimates per model type. We report the LOCO AUC drop as the difference between within-cohort (nested CV) AUC and LOCO AUC.

Pairwise transfer: A classifier was trained on one cohort and tested on each of the other three cohorts individually, yielding a 4×4 matrix (diagonal excluded) of cross-cohort AUC values per model type. This design is more stringent than LOCO and isolates specific directional failures.

Bootstrap 95% CIs (1,000 resamples) on the test set predictions are reported for all LOCO and pairwise estimates.

2.6 PERMANOVA Variance Decomposition

To quantify the relative contribution of cohort identity and disease status to total compositional variance, we conducted PERMANOVA (permutational multivariate analysis of variance) using the `adonis2` function in the R package `vegan` v2.6 (Oksanen et al., 2024). Two distance matrices were used:

- **Aitchison distance** (Euclidean distance on CLR values): the compositionally appropriate distance that does not distort due to the simplex constraint.
- **Bray-Curtis dissimilarity** (on relative abundances): included for comparability with the broader microbiome literature.

All PERMANOVA models used 9,999 permutations with Type III (marginal) sums of squares (by="margin") to allow independent assessment of cohort and diagnosis effects. Three models were fitted: (1) cohort only, all five cohorts; (2) cohort + diagnosis, restricted to the four labeled cohorts, with marginal R^2 for each term; (3) within each labeled cohort, diagnosis as the sole predictor (within-cohort disease effect size).

For Zhu 2022, analysis was performed on the 60 fecal samples only; the blood microbiome samples were excluded prior to all PERMANOVA analyses.

Homogeneity of multivariate dispersions was tested using `betadisper` and `permutest` (9,999 permutations) to assess whether observed cohort differences reflected centroid separation, within-group spread heterogeneity, or both. A significant beta-dispersion test indicates that PERMANOVA R^2 values partially reflect variation in within-group spread, not only centroid shifts.

2.7 Batch Correction

Two batch correction methods designed for microbiome data were applied, treating cohort identity as the batch variable and disease status as the biological covariate of interest:

ComBat-seq (Zhang et al., 2020): A negative binomial regression-based batch correction method adapted from the original ComBat algorithm for use with RNA-seq count data. We provided raw genus-level count matrices (genera × samples, four labeled cohorts only) and specified group = disease_status to preserve the biological variable. Corrected count tables were CLR-transformed using the same multiplicative zero-replacement procedure described in Section 2.3.

MMUPHin v1.15 (Ma et al., 2022): A meta-analysis framework for microbiome data that fits linear mixed-effects models to remove batch effects from relative abundance profiles while adjusting for covariates. Relative abundance profiles from the four labeled cohorts were provided, with batch = cohort and covariates = diagnosis.

After correction, the LOCO generalization experiment (Section 2.5) was repeated on the batch-corrected CLR matrices. We also re-computed within-cohort AUC on corrected data to assess whether correction degraded within-cohort biological signal. We note that because batch correction was applied to all four cohorts jointly—including the held-out test cohort—the corrected LOCO results represent an upper bound on what batch correction could achieve in a prospective deployment scenario where the test cohort would be unseen during correction parameter estimation.

2.8 SHAP Feature Importance Analysis

To identify which genera drove predictions in each cohort and whether these features were consistent across cohorts, we computed SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017).

For each labeled cohort, a final classifier was trained on all samples in that cohort (best hyperparameters from 5-fold cross-validation), and SHAP values were computed for all samples using `shap.LinearExplainer` (for logistic regression, `feature_perturbation="interventional"`) and `shap.TreeExplainer` (for LightGBM, `check_additivity=False`). The mean absolute SHAP value per genus was used as the importance metric; the mean SHAP value (signed) per genus within AD samples versus CN samples indicated direction of association.

For LOCO analyses, SHAP values were computed on the held-out test cohort from each LOCO-trained model, allowing direct assessment of which genera the cross-cohort model relied upon when applied to an unseen population.

Pairwise taxonomic overlap between cohorts was quantified using Jaccard similarity on the top- N SHAP genera ($N \in \{10, 20, 50\}$). A genus was classified as a “directional flip” if it appeared in the top-20 SHAP features of at least two cohorts but had opposite sign (i.e., positive mean SHAP in AD samples in one cohort; negative in another), constituting an empirical cross-population directional contradiction.

All analyses used `MIN_COHORTS=2` for directional flip detection and `TOP_N=20` as the primary overlap threshold; top-10 and top-50 are reported as sensitivity analyses.

2.9 AI Tool Disclosure

Initial manuscript drafts and computational scripts were prepared with minimal assistance from Claude (Anthropic). All analytical decisions, parameter choices, experimental design, statistical interpretations, and scientific conclusions were made by the author. All code and text were reviewed, validated, and substantially revised under the author’s direction.

2.10 Statistical Analysis Environment

All Python analyses used `scikit-learn` v1.5, `lightgbm` v4, `shap` v0.51, `pandas` v2.x, and `numpy` v1.x in a conda environment (Python 3.11). R analyses used `DADA2` v1.28, `vegan` v2.6, `sva`

v3.50 (ComBat-seq), and MMUPHin v1.15 in R v4.3. Figures were generated with matplotlib v3.8 at 300 DPI.

3. Results

3.1 Cohort Characteristics and Processing

Five cohorts comprising 599 deposited samples from three countries were assembled (Figure 1A; Table 1). After DADA2 processing and chimera removal, read retention rates varied substantially across cohorts (47.9%–78.3%; Supplementary Table S2), reflecting differences in sequencing platform, library preparation, and amplicon length. The Kazakhstan cohort (single-end, NovaSeq) had notably different quality dynamics compared to the paired-end MiSeq cohorts. SILVA v138.1 genus-level classification and cross-cohort union produced a unified feature matrix of 396 genera. The 90 blood microbiome (B_*) samples from Zhu 2022 were excluded prior to all analyses (Section 2.4), leaving a working dataset of 509 samples across five cohorts. CLR transformation was applied to all 509 samples jointly after union, with multiplicative zero-replacement at $\delta=0.65$.

For supervised classifier analyses (Phases 2–3 and 5–6), the Kim/KBASE 2022 cohort was excluded because per-sample diagnosis labels are IRB-restricted (Supplementary Note S1), leaving four labeled cohorts with a total of 401 samples for binary AD/CN analyses (86 Zhuang + 171 Ling + 60 Zhu fecal binary subset + 84 Kazakhstan; Zhu 2022 MCI samples excluded from binary analyses).

3.2 Within-Cohort Classification Performance

Within-cohort nested cross-validation yielded AUC-ROC values broadly consistent with those reported in the original publications (Figure 2; Table 2). Logistic regression achieved AUC-ROC of 0.633 [95% CI: 0.51–0.75] for Zhuang 2018, 0.865 [0.81–0.92] for Ling 2020, 0.998 [0.99–1.00] for Zhu 2022, and 0.766 [0.66–0.86] for Kazakhstan. LightGBM yielded similar values: 0.635 [0.52–0.76], 0.862 [0.81–0.91], 0.979 [0.95–1.00], and 0.694 [0.57–0.80] respectively.

The near-perfect within-cohort AUC for Zhu 2022 fecal samples (logistic regression 0.998; LightGBM 0.979; $n=60$ binary) is notable given the small sample size (30 per class). With 10-fold CV on 60 samples, each test fold contains approximately six samples; fold-level AUC estimates can be unstable at this scale. PERMANOVA analysis (Section 3.4) reveals that diagnosis explains $R^2=0.094$ of within-cohort compositional variance ($F=6.01$, $p<0.0001$), suggesting genuine group differences in fecal composition in this cohort. However, the high AUC should be interpreted cautiously and the LOCO experiment (Section 3.3) provides a more informative assessment of how transferable this signal is.

The three-class analysis (AD / MCI / CN) on all 90 Zhu 2022 fecal samples using logistic regression (one-vs-rest macro-AUC under nested cross-validation) yielded macro-AUC=0.761, confirming the multi-class discriminability of this cohort's fecal compositional data.

For Zhuang 2018, within-cohort AUC values (0.633–0.635) were modest, consistent with the small sample size and weak within-cohort diagnosis effect observed in PERMANOVA ($R^2=0.017$, $p=0.066$; Section 3.4).

Table 1. Cohort characteristics.

Cohort	Country	Accession	N (analyzed)	AD	CN	MCI	Sequencing	Library	Supervised Analysis
Zhuang 2018	China	PRJNA554111	86	43	43	—	MiSeq 2×300	Paired-end	Phases 2–6
Ling 2020	China	PRJNA633959	171	100	71	—	MiSeq 2×300	Paired-end	Phases 2–6
Zhu 2022†	China	PRJNA489760	90 (fecal)	30	30	30	MiSeq 2×300	Paired-end	Phases 2–6
Kaiyrlkyzy 2022	Kazakhstan	PRJNA811324	84	41	43	—	NovaSeq	Single-end	Phases 2–6
Kim 2022	South Korea	PRJEB50447	78	18*	60	—	MiSeq 2×300	Paired-end	Phase 4 only
Total			509	232	247	30			

Amyloid-PET positive preclinical AD; per-sample labels IRB-restricted and not publicly available. †Zhu 2022 deposited 180 samples (90 fecal F₋ + 90 blood B₋); only fecal samples retained for analysis.*

Table 2. Within-cohort nested cross-validation performance (10-fold outer, 5-fold inner CV).

Cohort	Model	N	AUC-ROC	95% CI	Sensitivity	Specificity
Zhuang 2018	LogReg	86	0.633	[0.51–0.75]	0.814	0.512
Zhuang 2018	LGBM	86	0.635	[0.52–0.76]	0.884	0.442
Ling 2020	LogReg	171	0.865	[0.81–0.92]	0.780	0.859
Ling 2020	LGBM	171	0.862	[0.81–0.91]	0.740	0.887
Zhu 2022	LogReg	60	0.998	[0.99–1.00]	0.967	1.000

Cohort	Model	N	AUC-ROC	95% CI	Sensitivity	Specificity
Zhu 2022	LGBM	60	0.979	[0.95–1.00]	0.900	1.000
Kaiyrlyk yzy 2022	LogReg	84	0.766	[0.66–0.86]	0.829	0.605
Kaiyrlyk yzy 2022	LGBM	84	0.694	[0.57–0.80]	0.610	0.767

N=60 for Zhu 2022 binary analysis (30 AD + 30 CN fecal samples; MCI and blood samples excluded).

3.3 Cross-Cohort Generalization Failure

LOCO cross-cohort AUC-ROC values revealed consistent and substantial generalization failure across all cohorts and model types (Figure 3A; Table 3). Mean LOCO AUC across the four cohorts was 0.640 for logistic regression and 0.647 for LightGBM—corresponding to mean drops of 0.176 and 0.146 AUC points respectively from within-cohort performance. One of eight model-cohort combinations produced LOCO AUC near chance: Zhuang 2018 with logistic regression (AUC=0.504 [0.37–0.63]).

The largest absolute logistic regression AUC drop was observed for Kazakhstan (within-cohort 0.766 → LOCO 0.564, $\Delta=0.202$). Zhu 2022 showed the largest LightGBM drop: within-cohort 0.979 → LOCO 0.758 ($\Delta=0.221$); its logistic regression drop was also substantial (0.998 → 0.813, $\Delta=0.185$). In both cases the LOCO model must rely on disease-relevant genera from other cohorts that lack the cohort-specific compositional patterns inflating within-cohort performance.

Ling 2020 also experienced substantial LOCO collapse: logistic regression 0.865 → 0.678 ($\Delta=0.187$); LightGBM 0.862 → 0.674 ($\Delta=0.188$). LightGBM confidence intervals for the Ling 2020 LOCO estimate [0.59–0.75] do not overlap with the within-cohort estimate [0.81–0.91], confirming a statistically reliable performance drop.

Table 3. LOCO cross-cohort AUC-ROC versus within-cohort nested CV baseline.

Test Cohort	Model	Within-Cohort AUC	LOCO AUC	LOCO 95% CI	AUC Drop
Zhuang 2018	LogReg	0.633	0.504	[0.37–0.63]	0.129
Zhuang 2018	LGBM	0.635	0.565	[0.44–0.69]	0.070
Ling 2020	LogReg	0.865	0.678	[0.59–0.76]	0.187
Ling 2020	LGBM	0.862	0.674	[0.59–0.75]	0.188
Zhu 2022	LogReg	0.998	0.813	[0.69–0.92]	0.185
Zhu 2022	LGBM	0.979	0.758	[0.63–0.87]	0.221
Kazakhstan	LogReg	0.766	0.564	[0.44–0.69]	0.202
Kazakhstan	LGBM	0.694	0.591	[0.47–0.71]	0.103
Mean	LogReg	0.816	0.640	—	0.176

Test Cohort	Model	Within-Cohort AUC	LOCO AUC	LOCO 95% CI	AUC Drop
Mean	LGBM	0.793	0.647	—	0.146

Pairwise single-cohort transfer AUC values (Figure 3B–C; Supplementary Table S1) ranged from 0.54 to 0.88 for logistic regression. The highest pairwise AUC was Ling 2020 → Zhu 2022 (logistic regression AUC=0.88), reflecting transfer between two large Chinese fecal cohorts with overlapping compositional structure. Kazakhstan → Ling 2020 also transferred reasonably (logistic regression AUC=0.74). Most other pairwise AUCs clustered between 0.54 and 0.70, consistent with the overall pattern of limited cross-cohort transferability.

3.4 PERMANOVA Variance Decomposition

PERMANOVA on Aitchison distance across all five cohorts showed that cohort identity explained 19.3% of total compositional variance ($R^2=0.193$, $F=30.13$, $p<0.0001$, 9,999 permutations; Figure 4A). On Bray-Curtis dissimilarity, the cohort R^2 was 0.136 ($F=19.80$, $p<0.0001$). The cohort effect was substantially larger than disease signal: in a marginal two-variable model restricted to the four labeled cohorts, cohort (marginal) explained $R^2=0.172$ ($F=28.03$, $p<0.0001$) versus diagnosis (marginal) $R^2=0.014$ ($F=6.82$, $p<0.0001$)—an approximately 12-fold disparity. This ratio is the single most important number for mechanistically interpreting the batch correction results in Section 3.5.

Principal coordinates analysis (PCoA) on Aitchison distance (Figure 4B) visually confirmed strong cohort-level clustering, with partial overlap between Chinese cohorts (Zhuang 2018, Ling 2020) but clear separation of Kazakhstan and Zhu 2022. The Kim/KBASE 2022 cohort was clearly separated from all others along PC1, consistent with its preclinical amyloid-based staging, South Korean origin, and potentially distinct dietary patterns.

Beta-dispersion testing revealed significant heterogeneity in within-group spread across cohorts ($F=13.93$, $p<0.0001$, 9,999 permutations; Figure 4C), meaning that the PERMANOVA R^2 of 0.193 reflects both compositional centroid shifts and differences in within-cohort spread. Kazakhstan showed the greatest within-cohort variance, while the Chinese cohorts (Zhuang, Ling) were more tightly dispersed. This dispersion heterogeneity is relevant to interpreting PERMANOVA effect sizes and is acknowledged in the Methods section.

Within-cohort PERMANOVA revealed highly variable disease-associated compositional effects: Ling 2020 ($R^2=0.053$, $F=9.53$, $p<0.0001$) and Zhu 2022 ($R^2=0.094$, $F=6.01$, $p<0.0001$) showed significant diagnosis effects, while Zhuang 2018 ($R^2=0.017$, $p=0.066$) and Kazakhstan ($R^2=0.022$, $p=0.069$) did not reach statistical significance at $\alpha=0.05$. These within-cohort disease effect sizes broadly rank-order with within-cohort AUC performance, providing a triangulating validation of the classifier results. The Zhu 2022 fecal-only $R^2=0.094$ represents the highest within-cohort disease effect among the four labeled cohorts, consistent with a genuine compositional signal in this dataset.

3.5 Batch Correction Does Not Recover Cross-Cohort Generalization

Batch correction was applied using ComBat-seq and MMUPHin to the four labeled cohorts with cohort as the batch variable (Figure 5; Table 4). Both methods failed to recover cross-cohort generalization, and ComBat-seq substantially worsened it.

Table 4. Mean LOCO AUC across four cohorts, by correction method and model.

Method	LogReg Mean LOCO AUC	LogReg Mean Drop from Within-Cohort	LGBM Mean LOCO AUC	LGBM Mean Drop
Uncorrected	0.640	0.176	0.647	0.146
ComBat-seq	0.493	0.322	0.555	0.238
MMUPHin	0.612	0.203	0.582	0.210

ComBat-seq reduced mean logistic regression LOCO AUC from 0.640 to 0.493 and degraded all four individual cohort LOCO AUCs, with the Kazakhstan cohort collapsing to AUC=0.308 [0.20–0.42] (Table 4). LightGBM was somewhat more robust: ComBat-seq reduced LGBM mean LOCO from 0.647 to 0.555. MMUPHin was less destructive but still failed to improve upon uncorrected LOCO performance (LogReg mean: 0.640 → 0.612; LGBM: 0.647 → 0.582).

Critically, within-cohort AUC on corrected data was largely preserved (within ± 0.05 of uncorrected) for most cohort–method combinations, ruling out the hypothesis that batch correction was simply removing all signal indiscriminately. The exception was Kazakhstan, where both correction methods moderately improved within-cohort AUC (logistic regression: 0.766 → 0.784 with ComBat-seq, 0.766 → 0.788 with MMUPHin), while simultaneously collapsing LOCO AUC to below chance. This pattern—improved within-cohort AUC coupled with worsened cross-cohort AUC—is consistent with the correction algorithm overfitting the batch-correction transformation to Kazakhstan’s within-cohort structure, making its representation less compatible with the other cohorts, not more.

The mechanistic interpretation follows directly from the PERMANOVA result: when cohort explains ~12 times more variance than disease status, any linear correction that aligns cohort distributions necessarily corrupts the disease axis. The correction cannot simultaneously remove cohort-associated variance and preserve the much smaller disease signal, because those two sources of variance are not separable by linear methods in a composition space dominated by cohort-level differences.

3.6 SHAP Analysis Reveals Cohort-Specific and Contradictory Taxonomic Signatures

To characterize the mechanistic basis of generalization failure, we computed SHAP values for within-cohort classifiers and LOCO-trained classifiers (Figure 6).

Minimal overlap in top predictive taxa across cohorts. Pairwise Jaccard similarity of the top-20 SHAP genera between cohort pairs averaged 0.101 for logistic regression and 0.026 for LightGBM (Figure 6C). These near-zero overlap values confirm that within-cohort classifiers are not detecting a shared biological signal—they are detecting distinct, population-specific

compositional patterns. The higher Jaccard for logistic regression (0.101 vs 0.026 for LightGBM) reflects the greater sparsity of tree model feature selection: LightGBM concentrates importance in fewer, more cohort-specific features.

Cohort-specific top predictors. The most discriminative genus within each cohort was highly variable: *Subdoligranulum* (CN-associated, Zhuang 2018, logistic regression; |SHAP|=0.320) versus *Akkermansia* (CN-associated, Ling 2020, logistic regression; |SHAP|=0.372) versus *Streptococcus* (CN-associated, Zhu 2022, logistic regression; AD-associated for LightGBM) versus *Akkermansia* (CN-associated, Kazakhstan, logistic regression; |SHAP|=0.132). *Akkermansia* ranked first in both Ling 2020 and Kazakhstan for logistic regression, though both associations were CN-directed. For LightGBM, the top taxon per cohort was entirely distinct: *Oscillibacter* (Zhuang 2018), *Akkermansia* (Ling 2020, AD-associated), *Streptococcus* (Zhu 2022, AD-associated), *Castellaniella* (Kazakhstan, CN-associated) (Figure 6A).

Directional flip taxa. Among taxa appearing in the top-20 for two or more cohorts, ten showed directional reversal for logistic regression and three for LightGBM (Figure 6B):

Logistic regression flips (10 taxa): Agathobacter (AD-associated in Zhuang/Ling/Kazakhstan; CN-associated in Zhu 2022), Bifidobacterium (AD-associated in Zhuang/Zhu 2022/Kazakhstan; CN-associated in Ling), Christensenellaceae R-7 group (CN-associated in Zhuang/Ling/Zhu 2022; AD-associated in Kazakhstan), Coprococcus (CN-associated in Zhuang/Ling/Kazakhstan; AD-associated in Zhu 2022), Lactobacillus (CN-associated in Chinese cohorts; AD-associated in Kazakhstan), NK4A214 group (CN-associated in Zhuang/Ling; AD-associated in Zhu 2022/Kazakhstan), Parasutterella (CN-associated in Zhuang/Kazakhstan; AD-associated in Ling/Zhu 2022), Prevotella_9 (CN-associated in Zhuang/Kazakhstan; AD-associated in Ling/Zhu 2022), Romboutsia (AD-associated in Zhuang/Kazakhstan; CN-associated in Ling/Zhu 2022), and *[Eubacterium] xylanophilum group* (CN-associated in Zhuang; AD-associated in Ling/Zhu 2022/Kazakhstan).

LightGBM flips (3 taxa): Incertae Sedis (CN-associated in Zhuang/Zhu 2022/Kazakhstan; AD-associated in Ling), Lactobacillus (CN-associated in Ling/Zhu 2022/Zhuang; AD-associated in Kazakhstan), Oscillibacter (CN-associated in Zhuang/Ling/Zhu 2022; AD-associated in Kazakhstan).

The *Akkermansia* flip is particularly striking: in Ling 2020, *Akkermansia* is the top LightGBM predictor (|SHAP|=0.828) and is AD-associated—yet it is CN-associated in the other three cohorts for LightGBM (and CN-associated in all four cohorts for logistic regression).

Akkermansia muciniphila has been extensively studied for its purported gut-protective effects, and its relationship with AD pathology has been reported in opposing directions across populations (Liu et al., 2020). Our data empirically confirm this cross-population directionality reversal.

LOCO SHAP analysis: no universal transferable taxon. When SHAP values were computed for LOCO-trained classifiers applied to held-out cohorts, the top predictors in the test cohorts were entirely different across held-out conditions: *Akkermansia* (held-out Zhuang 2018, logistic regression), *Eisenbergiella* (held-out Ling 2020, logistic regression), *Faecalibacterium* (held-out Zhu 2022, logistic regression), and *Cutibacterium* (held-out Kazakhstan, logistic regression). For LightGBM, the top taxa were *Akkermansia* (held-out Zhuang), *Streptococcus* (held-out Ling), *Akkermansia* (held-out Zhu 2022), and *[Eubacterium] xylanophilum group* (held-out

Kazakhstan). This pattern indicates that the cross-cohort classifier applies cohort-inappropriate feature weights to the test cohort—it activates whichever genera most closely match its training-cohort signature, even when those genera lack consistent disease associations in the test population.

4. Discussion

4.1 Systematic Quantification of Generalization Failure

This study provides the first systematic, quantitative analysis of cross-cohort generalization failure in gut microbiome-based AD classifiers. Across four labeled cohorts, two model types, two cross-cohort evaluation designs, and two batch correction methods, we consistently observe the same result: classifiers trained within one population fail to generalize reliably to another. The mean LOCO AUC drop of 0.176 points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations near chance, demonstrates that previously reported within-cohort performance figures cannot be taken as evidence of a generalizable diagnostic signal.

This finding is not surprising in light of the broader literature. The *Bacteroides* directionality reversal documented by Jemimah et al. (2023) across North American and Chinese cohorts, the contradictory reports for *Akkermansia* (Liu et al., 2020), and the highly cohort-specific patterns reported in individual papers all suggest population-level heterogeneity. What this paper adds is a rigorous, unified quantification within a single computational framework, making the magnitude of failure explicit rather than anecdotal.

4.2 Why Batch Correction Failed: The 12:1 Disparity

The batch correction failure is mechanistically interpretable from the PERMANOVA result. Cohort identity explains 19.3% of Aitchison variance ($R^2=0.193$); disease status explains 1.4% ($R^2=0.014$). This ~12-fold disparity means that compositional space is overwhelmingly organized by cohort-of-origin, not by disease status. A linear batch correction method that targets the cohort dimension of variance will inevitably remove a large fraction of the disease variance, because those dimensions are not cleanly orthogonal in this data.

ComBat-seq’s aggressive worsening of LOCO AUC (mean LogReg: 0.640 $\rightarrow$ 0.493) supports this interpretation. ComBat-seq models each feature’s batch effect separately using a negative binomial regression framework that is well-suited to RNA-seq data but may be poorly calibrated for the extreme sparsity (>77% zero counts) and compositional structure of 16S microbiome data. The within-cohort AUC preservation (and even improvement for Kazakhstan) after ComBat-seq correction shows that the correction did not remove all biological signal—it reshaped the compositional space in a way that made within-cohort structure more accessible to linear classifiers while making cross-cohort alignment worse. This is consistent with ComBat-seq fitting a separate model per feature (per genus) without any constraint that the corrected compositions remain compatible across cohorts at the distributional level.

MMUPHin’s milder but still negative effect is similarly interpretable: linear mixed-effects models on relative abundances cannot align the full multivariate compositional space when the

batch effect is substantially larger than the biological effect. The corrected distributions may appear visually closer in PCoA (as shown in Supplementary Figures S4–S5) without actually aligning along the disease axis.

These results have direct implications for the microbiome harmonization literature. Recent methodological advances in microbiome batch correction have focused on cases where the biological effect is of similar magnitude to or larger than the batch effect. When the batch effect dominates by a factor of ~ 12 , no linear correction method—however sophisticated—can reliably separate the two sources of variance. The field may need fundamentally different approaches, such as multi-cohort training with cohort as a stratification variable, domain adaptation methods, or meta-analytic frameworks that explicitly model between-cohort heterogeneity rather than attempting to remove it.

4.3 Implications of the Zhu 2022 Paired-Sample Design

The Zhu 2022 cohort deposited both blood microbiome (B_{-}) and fecal (F_{-}) samples from the same 90 participants (PRJNA489760). The blood samples were excluded from this analysis because blood and fecal microbiome communities represent fundamentally distinct compartments; including both would introduce a sample-type confound identical in structure to a batch effect, inflating within-cohort AUC through tissue-compartment discrimination rather than disease discrimination.

With fecal-only samples ($n=60$ binary), the near-perfect within-cohort AUC (logistic regression 0.998, LightGBM 0.979) is more difficult to attribute to a single confound. The small sample size (30 per class) is relevant: with 10-fold CV on 60 samples, each test fold contains approximately 6 samples, and single-fold AUC estimates can be highly variable. PERMANOVA diagnosis $R^2=0.094$ ($F=6.01$, $p<0.0001$) is genuine—Zhu 2022 shows the highest within-cohort disease signal of the four labeled cohorts—but an AUC of 0.998 likely still reflects optimistic estimation at this sample size.

The LOCO result for Zhu 2022 (logistic regression AUC=0.813, the highest LOCO value among all cohorts) is informative: classifiers trained on the other three cohorts transfer to Zhu 2022 fecal samples better than to any other cohort. This suggests that the fecal disease signal in this cohort is somewhat consistent with signals in the Chinese and Kazakh training cohorts, unlike the spurious biopsy–fecal signal that would have characterized the original mixed dataset. We recommend that future AD microbiome studies report sample types explicitly and analyze each compartment independently before pooling.

4.4 Directional Flip Taxa: Empirical Confirmation of Cross-Population Contradictions

The directional flip taxa identified (ten for logistic regression, three for LightGBM) confirm empirically what the meta-analytic literature has suggested qualitatively. *Akkermansia* is the most prominent example: it is the top LightGBM predictor in Ling 2020 (AD-associated, |SHAP|=0.828) yet consistently CN-associated in Zhuang 2018, Zhu 2022, and Kazakhstan for LightGBM—and CN-associated in all four cohorts for logistic regression. *A. muciniphila* abundance is strongly influenced by host diet, particularly fiber and polyphenol intake (Plovier et al., 2017), and by geographic-population-specific dietary patterns. Chinese and Kazakh diets

differ substantially in fermented food consumption, fiber sources, and protein composition; these dietary differences likely modulate *Akkermansia* abundance independently of any AD-specific effect, producing the observed directional reversal.

Lactobacillus shows the opposite pattern: CN-associated in Chinese cohorts (Ling 2020, Zhu 2022) but AD-associated in Kazakhstan. Fermented dairy consumption (kumis, kefir) is culturally prominent in Central Asian populations and would be expected to substantially elevate *Lactobacillus* abundance in Kazakh controls—and by extension, in healthy older Kazakh participants. If healthy Kazakh controls have higher *Lactobacillus* due to diet, the taxon would be CN-associated in that cohort; if the AD patients in that cohort do not consume fermented dairy at the same rate, the direction would reverse relative to Chinese cohorts where fermented dairy consumption patterns differ.

These diet-mediated directional reversals have a critical implication: they are not correctable by standard batch correction methods, because they reflect genuine biological differences in the meaning of a taxon's abundance—not technical artifacts of sequencing. The same relative abundance of *Akkermansia* carries different disease-predictive information in a Chinese urban cohort versus a Kazakh cohort, because the baseline compositional ecology differs. This is a biological generalization problem, not a technical harmonization problem.

4.5 Implications for Clinical Translation

The results of this study suggest that gut microbiome-based AD classifiers are not ready for cross-population clinical application in their current form. A classifier trained on a Chinese cohort and applied to a Kazakh patient population would perform at or below chance for logistic regression, and only modestly above chance even for LightGBM. These performance levels are clinically inadequate for any diagnostic application.

We propose three prerequisites for meaningful clinical translation of microbiome-based AD diagnostics: (1) **Multi-cohort harmonized training data:** classifiers should be trained on samples from multiple geographic and demographic populations simultaneously, with cohort or population as a stratification variable, rather than training within a single cohort and expecting transfer. (2) **Prospective cross-cohort validation as a publication standard:** any future study reporting a gut microbiome AD classifier should be required to validate performance on at least one independent cohort from a different geographic population before claiming clinical relevance. The within-cohort AUC, no matter how high, is an insufficient evidential standard. (3) **Covariate-adjusted models:** diet, geographic origin, antibiotic history, and other known microbiome-modulating covariates should be incorporated into classification models either as features or confounders, to reduce the influence of population-specific compositional baselines.

The multi-cohort harmonized collection approach has succeeded in other biomarker domains (e.g., the UK Biobank for genomic predictors, the ADNI consortium for neuroimaging). A parallel initiative—perhaps a consortium of AD microbiome studies committed to harmonized collection protocols and prospective cross-cohort sharing—would be the most direct route to a clinically actionable microbiome diagnostic.

4.6 Limitations

Several limitations of this study should be acknowledged explicitly.

Mixed single-end and paired-end sequencing: Kazakhstan used single-end NovaSeq reads while all other cohorts used paired-end MiSeq 2×300 bp. Single-end reads yield lower taxonomic resolution at equivalent read length due to the absence of overlap-based error correction in DADA2, and different error models were fitted per cohort. This technical asymmetry contributes to between-cohort heterogeneity and cannot be disentangled from biological population differences in this dataset.

Kim/KBASE 2022 excluded from supervised analyses: Per-sample amyloid-PET diagnosis labels are IRB-restricted and not publicly deposited. The Kim 2022 cohort (78 participants; 18 amyloid-positive preclinical AD, 60 CN) therefore contributed only to PERMANOVA variance decomposition. The use of amyloid-PET positivity as the outcome (preclinical AD) rather than clinical diagnosis also means it is not directly comparable to the other four cohorts even if labels were available.

China overrepresentation: Three of the four labeled cohorts are Chinese. The cross-cohort experiment is therefore primarily a China-internal transfer (Zhuang ↔ Ling ↔ Zhu 2022) plus transfer to/from a single Central Asian cohort. A more geographically balanced design—incorporating North American, European, and African populations—would be needed to generalize the conclusions to the full range of human diversity.

Partial SRA deposit for Zhu 2022: Only 180 of the 302 originally enrolled participants were deposited in the public archive (PRJNA489760). The deposited subset may not be a random sample of the full study population, potentially introducing selection bias of unknown direction.

Batch correction leakage in LOCO evaluation: The batch correction parameters (ComBat-seq’s per-gene normalization factors; MMUPHin’s batch effect estimates) were estimated using all four labeled cohorts jointly, including the LOCO test cohort. In a prospective deployment scenario, the test cohort would be unseen during correction. The corrected LOCO AUC values reported here therefore represent an upper bound on corrected performance; true prospective performance would likely be worse.

Cross-sectional design and observational data: All cohorts used cross-sectional sampling. Causal inference about microbiome–AD relationships is not supported by this design. All language throughout this paper refers to associations and predictive performance, not causal effects.

5. Conclusion

We conducted the first systematic cross-cohort generalization evaluation of gut microbiome-based Alzheimer’s disease classifiers, spanning five independent cohorts from three countries. Within-cohort classifiers achieved AUC-ROC values of 0.633–0.998, consistent with published reports; however, LOCO cross-cohort AUC values fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression. Standard batch correction methods (ComBat-seq and MMUPHin) worsened rather than improved cross-cohort AUC. PERMANOVA revealed that cohort identity explained approximately 12 times more compositional variance than disease status, providing a mechanistic explanation for why batch correction fails. SHAP analysis identified near-zero taxon overlap between cohort-specific predictive signatures (mean Jaccard

0.101 at top-20 for logistic regression) and ten directional flip taxa for logistic regression and three for LightGBM—genera whose AD or CN association reverses across populations. These findings demonstrate that the gut microbiome-based AD classifier literature has a fundamental generalization problem that cannot be resolved by post-hoc batch correction. Multi-cohort harmonized prospective studies and mandatory cross-cohort validation are needed before microbiome-based AD biomarkers can achieve clinical utility.

Data Availability Statement

All raw sequencing data analyzed in this study are publicly available from NCBI SRA or ENA under accession numbers PRJNA554111 (Zhuang 2018), PRJNA633959 (Ling 2020), PRJNA489760 (Zhu 2022), PRJNA811324 (Kaiyrlykyzy 2022), and PRJEB50447 (Kim 2022). Analysis code, processed data tables, and figure generation scripts are available at <https://github.com/carhanc/microbiome-ad-generalization>.

Author Contributions

A.C. conceived the study, developed and executed the complete analysis pipeline, interpreted the results, and wrote the manuscript.

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Conflict of Interest

The author declares no conflict of interest.

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